

WinPE Autopilot / Intune Diagnostic Tool

User Manual

For Authorised Technicians

Tool Version	1.0
Platform	Windows PE (WinPE)
Supported OS	Windows 10 / 11 (offline)
Document Status	Confidential

1. Overview

This tool is a Windows PE-based diagnostic utility designed for PC rental and resale operations. It scans a device's installed Windows drive — while the OS is completely offline — and checks whether any Microsoft Intune or Autopilot registration traces remain on that device before it is returned to inventory, re-rented to a new client, or sold to a dealer.

Running this check before each device handover helps your organization avoid two common and costly issues:

- **Re-rental risk:** A returning device that was enrolled in a previous client's Intune/Autopilot profile may still appear linked to that client's management system. Deploying it to a new client without proper cleanup creates a serious security and compliance violation.
- **Resale risk:** Used PC dealers routinely inspect devices for residual MDM (Mobile Device Management) metadata. If such metadata is found, the assessed resale price is significantly reduced. This tool enables your team to detect and remediate the issue before the device reaches the dealer.

2. What the Tool Checks

2.1 Intune Status Check

The tool reads the device's offline registry and verifies whether the device is genuinely unregistered from Microsoft Intune. It examines:

- The Enrolments registry key (SOFTWARE\Microsoft\Enrolments)
- MDM enrolment certificates in the system certificate store
- The DiscoveryServiceServerUrl value, which points to manage.microsoft.com when active enrolment exists

A device is reported as Genuinely Unregistered only when all of the above checks return clean. A single positive finding is sufficient to flag the device as NOT genuinely unregistered.

2.2 Autopilot Trace Check

The tool scans multiple locations for residual Autopilot registration data:

- Autopilot Diagnostics registry key (SOFTWARE\Microsoft\Provisioning\Diagnostics\Autopilot)
- AutoPilot configuration file on disk (AutoPilotConfigurationFile.json)
- CSP policy data (SOFTWARE\Microsoft\PolicyManager\current\device\Autopilot)
- Residual Autopilot metadata (SOFTWARE\Microsoft\Windows\Autopilot)

2.3 Graph API Tenant Verification (Optional)

If Autopilot traces are found locally, the tool can optionally connect to Microsoft's Graph API to cross-verify whether the device's serial number is still actively registered in a tenant's Autopilot deployment. This provides an authoritative, server-side confirmation beyond local registry checks.

NOTE: *This feature requires network connectivity and valid tenant credentials (Tenant ID, Client ID, Client Secret). Without these, the tool still performs the full local scan and generates a report.*

2.4 Hardware Information

The tool also collects the following device information for reporting purposes:

- Computer name (from the offline OS registry)
- Laptop model name (from BIOS/SMBIOS via wmic)
- Serial number (from BIOS/SMBIOS)
- Hardware UUID
- TPM availability
- Azure AD / Entra ID join status

3. Prerequisites

3.1 What You Need

Boot USB	A Windows Installation USB (created with the Microsoft Media Creation Tool)
Tool USB	A second USB drive containing test.exe
Target device	The PC or laptop to be scanned (powered off before you begin)
Network (optional)	Required only for the Graph API tenant verification step

3.2 USB Drive Layout

The tool USB should contain the following structure:

```
E:\  
  test.exe  
  Diagnostics\  
    Results\      (created automatically — reports are saved  
here)
```

NOTE: The Results folder is created automatically on first run. Do not manually place files inside it.

(Here, E: is the example tool USB drive; the users' directory can be different.)

4. Step-by-Step Usage

Step 1 — Prepare the Target Device

- Ensure the device is powered off completely.
- Do NOT boot into the device's installed Windows before scanning. The scan must run against an offline OS.

Step 2 — Insert Both USB Drives

- Insert the Windows Installation USB (this is the boot USB).
- Insert the Tool USB (containing test.exe).

Step 3 — Boot into WinPE

- Power on the device.

- Press the Boot Menu key immediately when the manufacturer logo appears. Common keys:

HP	F9
Dell	F12
Lenovo	F12
Asus	Esc or F8
Acer	F12

- Select the Windows Installation USB from the boot menu.
- When the Windows Setup screen appears, press Shift + F10 to open a Command Prompt. Do NOT click "Install now".

WARNING: Clicking "Install now" will start a Windows installation. Only press Shift + F10 to open the command prompt.

Step 4 — Locate the Tool USB

In the Command Prompt, identify the drive letter assigned to your Tool USB:

```
diskpart
list volume
exit
```

Look for a Removable volume that matches your Tool USB size. The Boot Installation USB will appear as ESD-USB or BOOTFS. Your Tool USB is the other removable drive.

Step 5 — Assign a Drive Letter to the Target OS (if needed)

The target device's internal Windows drive may not have a drive letter assigned in WinPE. Check the volume list from Step 4. If the internal NTFS partition (typically the largest volume) shows no letter in the Ltr column, assign one:

```
diskpart
select volume 2          (use the correct volume number from the list)
assign letter=E
exit
```

NOTE: The drive letter assignment is temporary and only lasts for this WinPE session.

Step 6 — Run the Tool

Navigate to the Tool USB and run the executable:

```
E:
test
```

(Here E: is the example tool USB drive; the users' directory can be different.)

The tool will automatically:

- Detect the target OS drive (the installed Windows, not WinPE itself)
- Load the offline registry hives

- Perform all local Intune and Autopilot checks
- Collect hardware information (computer name, model, serial number, UUID)

Step 7 — Graph API Check (Optional)

If Autopilot traces are detected locally, the tool will prompt you to enter tenant credentials for the Graph API verification step:

```
--- Graph API / Tenant Configuration ---
(Press Enter on Tenant ID to skip Graph API check)
Tenant ID      : [enter or press Enter to skip]
Client ID      : [enter]
Client Secret  : [enter]
Token Base URL : [leave blank for real Microsoft endpoint]
Graph Base URL : [leave blank for real Microsoft endpoint]
```

If you press Enter without typing a Tenant ID, the Graph API step is skipped and the tool proceeds with the local scan results only.

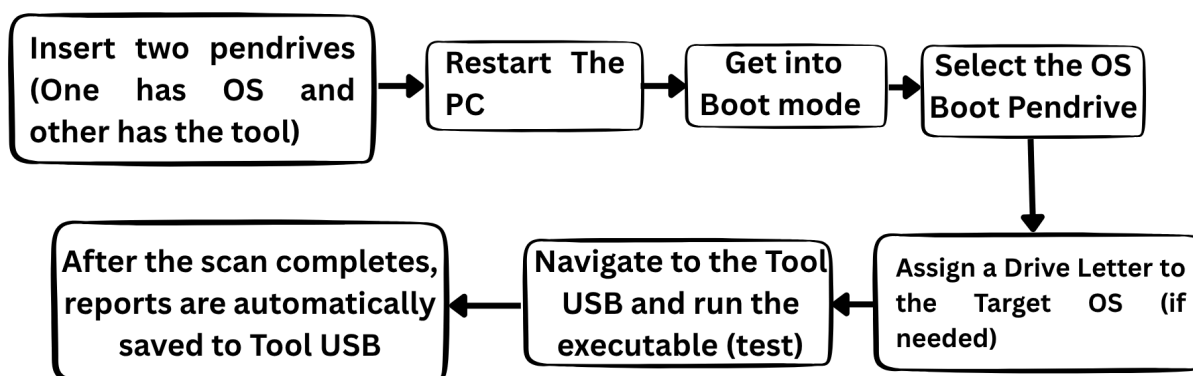
Step 8 — View and Retrieve Reports

After the scan completes, reports are automatically saved to your Tool USB:

```
E:\Diagnostics\Results\report_<serial>_<timestamp>.json
E:\Diagnostics\Results\report_<serial>_<timestamp>.html
E:\Diagnostics\Results\report_<serial>_<timestamp>.csv
```

The HTML report can be opened in any web browser for easy reading. The CSV is suitable for importing into Excel or a database for record-keeping.

WARNING: Reports are saved to the Tool USB, not to the WinPE RAM disk (X:). Always use the Tool USB so reports survive after the session ends.



System Execution Block Diagram

5. Understanding the Report

5.1 Risk Level

Every scan produces one of two verdicts:

CLEAN	No Autopilot or Intune traces found. The device is genuinely unregistered and safe for re-rental or resale.
RISK	One or more Autopilot or Intune traces were found. The device must be remediated before it is handed over to a new client or sold.

5.2 Report Fields

Computer Name	The hostname stored in the device's offline Windows registry
Model	The hardware model name reported by BIOS/SMBIOS (e.g. HP EliteBook 840 G8)
Serial Number	The hardware serial number from BIOS/SMBIOS
Serial Reliable	"true" if the serial was obtained directly from BIOS; "false" if unavailable
UUID	The hardware UUID of the device
TPM Available	"true" if a TPM chip is detected on the device
Azure AD Joined	"true" if the device was joined to Entra ID / Azure Active Directory
Intune Status Check	"Genuinely Unregistered" or "NOT Genuinely Unregistered (traces found)"
Autopilot Registry	"true" if Autopilot diagnostic registry keys are present
Enrollment Registry	"true" if Intune enrollment registry keys are present
Intune Enrolled	"true" if an active MDM enrollment URL pointing to manage.microsoft.com was found
MDM Certificate	"true" if certificates are present in the MDM certificate store
Tenant Status	Result of the Graph API check: "Still Registered", "Not Registered", or "Not Checked"

5.3 Tenant Status Values

Not Checked (Clean Locally)	No local Autopilot traces found; Graph API was not queried (no need to)
Cannot Verify — Missing Config	Autopilot traces found locally but no tenant credentials were entered
Not Registered	Graph API confirmed the serial number is NOT in the tenant's Autopilot registrations
Still Registered	Graph API confirmed the device IS still registered in the tenant — immediate action required
Error / Unverified	The Graph API call failed (network issue, wrong credentials, etc.)

6. Remediation Actions

When a device is flagged as RISK, the report will list the specific actions required. The table below summarises common remediation steps:

Intune traces found	Log in to the Microsoft Intune admin portal (intune.microsoft.com) and delete the device object from the Devices section.
Autopilot traces found	Log in to the Intune admin portal, go to Devices → Enrollment → Windows Autopilot Devices, locate the device by serial number, and delete it.
Graph API: Still Registered	The device is confirmed live in the tenant. Both Intune deletion and Autopilot deregistration must be completed before the device is cleared.

NOTE: After completing remediation steps, re-run this tool to confirm the device now reports as *CLEAN* before handing it over.

7. Troubleshooting

FATAL: Could not locate an offline target OS drive	The internal hard drive was not detected by WinPE. This is common with NVMe SSDs on some laptops. Try running diskpart → list disk to confirm the drive is visible. If not, the WinPE image may be missing the required storage driver.
FATAL: could not mount offline hives	The tool cannot access the registry hive files. Ensure you have completed Step 5 (drive letter assignment) and that the correct volume was selected.
Graph API check failed: token request failed	Network connectivity is unavailable or the credentials are incorrect. Verify the Tenant ID, Client ID, and Client Secret. Skip the Graph API step and rely on local scan results if network access is not available.
No report files in E:\Diagnostics\Results\	The scan may have failed silently. Re-run the tool and redirect output: test.exe > E:\out.txt 2>&1, then type E:\out.txt to read the full output.
list volume shows no internal drive	The storage controller driver is missing from the generic WinPE image. A custom WinPE image with the appropriate NVMe or RAID driver is required for this laptop model.

8. Security Notes

- This tool reads device data from an offline Windows installation. It does not modify any files on the target device.
- Tenant credentials (Client ID and Client Secret) are entered interactively and are never written to disk. Do not share them with unauthorized personnel.
- Report files contain device serial numbers and registration status. Store and transmit them securely and in accordance with your organization's data handling policy.
- This tool should only be operated by authorized technicians who understand the implications of the scan results.

9. Quick Reference

Boot Key (most HP)	F9
Boot Key (most Dell)	F12
Boot Key (most Lenovo)	F12
Open WinPE cmd	Shift + F10 on the Windows Setup screen
List volumes	diskpart → list volume → exit
Assign drive letter	diskpart → select volume N → assign letter=E → exit
Run the tool	E: then test.exe
Run with output log	test.exe > E:\out.txt 2>&1
Reports location	E:\Diagnostics\Results\
CLEAN result	Safe to re-rent or resell
RISK result	Remediation in Intune/Autopilot portal required before handover

9. Quick Flow

